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Practical week-1

Query 1: Create the database

The screenshot shows the SQL Developer interface. The Navigator pane on the left displays the SCHEMAS tree with 'pweek1' highlighted. The main editor pane shows a single SQL statement: 'create database pweek1'. The Output pane at the bottom shows the execution results.

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec

Query 2: Select the database

The screenshot shows the SQL Developer interface. The Navigator pane on the left displays the SCHEMAS tree with 'pweek1' highlighted. The main editor pane shows two SQL statements: 'create database pweek1' and 'use pweek1'. The Output pane at the bottom shows the execution results for both statements.

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec

Query 3: Create the books table

The screenshot shows the SQL Developer interface with the 'SCHEMAS' pane on the left and the 'SQL File 4*' editor in the center. The editor contains the following SQL code:

```
1 • create database pweek1
2 use pweek1
3
4 CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
✓ 1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
✓ 2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
✓ 3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec

Query 4: Create the members table

The screenshot shows the SQL Developer interface with the 'SCHEMAS' pane on the left and the 'SQL File 4*' editor in the center. The editor contains the following SQL code:

```
3
4 CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14
15 • CREATE TABLE members (
16     member_id INT AUTO_INCREMENT PRIMARY KEY,
17     full_name VARCHAR(100) NOT NULL,
18     email
19     VARCHAR(150) NOT NULL UNIQUE
20 );
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
✓ 1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
✓ 2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
✓ 3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
✓ 4	15:14:56	CREATE TABLE members (member_id INT AUTO_INC...	0 row(s) affected	0.079 sec

Query 5 (verification): Inspect the table structures

Navigator: SCHEMAS
Filter objects
joins
klu
pweek2
sys

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

My Snippets

```

13 );
14
15 CREATE TABLE members (
16     member_id INT AUTO_INCREMENT PRIMARY KEY,
17     full_name VARCHAR(100) NOT NULL,
18     email
19     VARCHAR(150) NOT NULL UNIQUE
20 );
21
22 describe books;

```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

Administration Schemas

Information

No object selected

Result 1 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
✓ 2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
✓ 3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
✓ 4	15:14:56	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected	0.079 sec
✓ 5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec

Query 6: Insert 3 books

Navigator: SCHEMAS
Filter objects
joins
klu
pweek2
sys

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

My Snippets

```

10 VARCHAR(13) NOT NULL UNIQUE,
11 published_year INT,
12 CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14
15 CREATE TABLE members (
16     member_id INT AUTO_INCREMENT PRIMARY KEY,
17     full_name VARCHAR(100) NOT NULL,
18     email
19     VARCHAR(150) NOT NULL UNIQUE
20 );
21
22 describe books;
23
24 INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);

```

Administration Schemas

Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
✓ 2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
✓ 3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
✓ 4	15:14:56	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected	0.079 sec
✓ 5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
✓ 6	15:20:03	INSERT INTO books (title, isbn, published_year) VALU...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

Query 7: Insert 3 members

Navigator

SQL File 3* SQL File 3* SQL File 4*

Limit to 1000 rows

SQLAdditions

My Snippets

SCHEMAS

Filter objects

joins

klu

pweek2

sys

Administration Schemas

Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:11:22	create database pweek1	1 row(s) affected	0.016 sec
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
3	15:14:08	CREATE TABLE books (book_id INT AUTO_INC...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VALU...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (A...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec

Object Info Session

Test 1: Duplicate ISBN (UNIQUE violation)

Navigator

SCHEMAS

Filter objects

- joins
- klu
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

SQLAdditions

My Snippets

```

18 email
19 VARCHAR(150) NOT NULL UNIQUE
20 );
21
22 • describe books;
23
24 • INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	15:13:25	use pweek1	0 row(s) affected	0.000 sec
3	15:14:08	CREATE TABLE books (book_id INT AUTO_I...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec

Object Info Session

Test 2: NULL title (NOT NULL violation)

Navigator

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

SQLAdditions

My Snippets

```

21
22 • describe books;
23
24 • INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);
36
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES (NULL, '9999999999999', 2010);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	15:14:08	CREATE TABLE books (book_id INT AUTO_I...	0 row(s) affected	0.109 sec
4	15:14:56	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec
9	15:24:36	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1048. Column 'title' cannot be null	0.000 sec

Object Info Session

Test 3: Future publication year (CHECK violation)

Navigator: SCHEMAS

Filter objects

- joins
- klu
- pweek1
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

My Snippets

```

24 • INSERT INTO books (title, isbn, published_year) VALUES
25 ('The Alchemist', '9780061122415', 1988),
26 ('Clean Code', '9780132350884', 2008),
27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);
36
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES (NULL, '999999999999', 2010);
39
40 • INSERT INTO books (title, isbn, published_year)
41 VALUES ('Time Traveler', '888888888888', 2030);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	15:14:56	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec
9	15:24:36	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1048. Column 'title' cannot be null	0.000 sec
10	15:25:18	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 3819. Check constraint 'chk_published_y...	0.000 sec

Object Info Session

Test 4: Duplicate email (UNIQUE violation)

Navigator: SCHEMAS

Filter objects

- joins
- klu
- pweek1
- pweek2
- sys

Administration Schemas

Information

No object selected

SQL File 3* SQL File 3* SQL File 4* x

Limit to 1000 rows

My Snippets

```

27 ('Atomic Habits', '9780735211292', 2018);
28
29 • INSERT INTO members (full_name, email) VALUES
30 ('Anil Kumar', 'anil.kumar@example.com'),
31 ('Priya Sharma', 'priya.sharma@example.com'),
32 ('Ravi Verma', 'ravi.verma@example.com');
33
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Fake Copy', '9780061122415', 2000);
36
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES (NULL, '999999999999', 2010);
39
40 • INSERT INTO books (title, isbn, published_year)
41 VALUES ('Time Traveler', '888888888888', 2030);
42
43 • INSERT INTO members (full_name, email)
44 VALUES ('Anil Clone', 'anil.kumar@example.com');

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	15:18:47	describe books	4 row(s) returned	0.016 sec / 0.000 sec
6	15:20:03	INSERT INTO books (title, isbn, published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
7	15:20:44	INSERT INTO members (full_name, email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
8	15:23:22	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1062. Duplicate entry '9780061122415' fo...	0.000 sec
9	15:24:36	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 1048. Column 'title' cannot be null	0.000 sec
10	15:25:18	INSERT INTO books (title, isbn, published_year) VAL...	Error Code: 3819. Check constraint 'chk_published_y...	0.000 sec
11	15:26:06	INSERT INTO members (full_name, email) VALUES (...	Error Code: 1062. Duplicate entry 'anil.kumar@exampl...	0.032 sec